Name: _____

WORKSHEET

Complex conjugates

1 Write the complex conjugate of each complex number

a $7 + 2i$

b $5 - 3i$

c $6 - 7i$

d $-4 - i$

e $3 + 10i$

f $5i - 6$

2 Find the sums of each of the following numbers and their conjugates

a $2 + 4i$

b $-6 + 9i$

c $1 - 5i$

d $-10 - 10i$

e $8 - 2i$

f $3i - 7$

3 Find the difference between each of the following numbers and their conjugates.

a $13 + 6i$

b $-9 - 3i$

c $-8 + i$

d $5 - 6i$

e $4i - 8$

f $-7 - 4i$

4 Find the product of each of the following numbers and their conjugates.

a $3 + i$

b $11 - 5i$

c $-5 + 4i$

d $-8 - 2i$

e $6i - 7$

f $10 - i$

5 For each of the following quadratic equations:

i Complete the square to give the difference of two squares where one square is imaginary

ii Find the complex conjugates that solve the equations.

iii Solve the equation.

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a $x^2 + 25 = 0$

b $3x^2 + 27 = 0$

c $(x + 2)^2 + 4 = 0$

d $(4x - 15)^2 = -16$

e $x^2 + 6x + 13 = 0$

f $x^2 - 10x = -61$

6 Use the quadratic formula to solve the following equations.

a $x^2 + 4x + 7 = 0$

b $x^2 - 6x + 14 = 0$

c $2x^2 + x + 1 = 0$

d $\frac{1}{2}x^2 - 5x = -17$

e $3x^2 = 7x - 7$

f $4x = 2x^2 + 6$

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Answers

1 **a** $7 - 2i$

b $5 + 3i$

c $6 + 7i$

d $-4 + i$

e $3 - 10i$

f $-5i - 6$

2 **a** 4

b -12

c 2

d -20

e 16

f -14

3 **a** $12i$

b $-6i$

c $2i$

d $-12i$

e $8i$

f $-8i$

4 **a** 10

b 146

c 41

d 68

e 85

f 101

5 **a** **i** $x^2 - (5i)^2$

ii $(x - 5i)(x + 5i)$

iii $x = \pm 5i$

b **i** $x^2 - (3i)^2$

ii $(x - 3i)(x + 3i)$

iii $x = \pm 3i$

c **i** $(x + 2)^2 - (2i)^2$

ii $(x + 2 + 2i)(x + 2 - 2i)$

iii $x = -2 \pm 2i$

d **i** $(4x - 15)^2 - (4i)^2$

ii $(4x - 15 + 4i)(4x - 15 - 4i)$

iii $x = \frac{15}{4} \pm i$

e **i** $(x + 3)^2 - (2i)^2$

ii $(x + 3 + 2i)(x + 3 - 2i)$

iii $x = -3 \pm 2i$

f **i** $(x - 5)^2 - (6i)^2$

ii $(x - 5 + 6i)(x - 5 - 6i)$

iii $x = 5 \pm 6i$

6 **a** $-2 \pm \sqrt{3}i$

b $3 \pm \sqrt{5}i$

c $-\frac{1}{4} \pm \frac{\sqrt{7}}{4}i$

d $5 \pm 3i$

e $\frac{7}{6} \pm \frac{\sqrt{35}}{6}i$

f $1 \pm \sqrt{2}i$

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